SECURELINK VPN LAB

Research Project | Secure VPN for Remote Access

Performance & Scalability Report

Establishing a Secure VPN for Remote Access

Deliverable D5 of 6

Organisation	Research project (lab build)
Author	Kshitij Rajesh Randhire
Version	1.0 (final)
Basis	Lab build on VMware; evidence IDs [E01]-[E38] refer to the evidence screenshot set
Status	Lab documentation. Not certified and not presented as production-ready.

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1 Method

iperf3 was run from the Kali client against gw-hq, with and without the VPN. The host is a shared Windows 10 machine (16 GB RAM) running all VMs on a virtual network, so the numbers show relative behaviour in this lab only. The load was one client, not many users.

2 Results

Run	Duration	No VPN	VPN 1 stream	VPN 5 streams	VPN 10 streams
Run 1 [E20]	8 s	3.40 Gbit/s	196 Mbit/s	151 Mbit/s	173 Mbit/s
Run 2 [E15]	10 s	1.90 Gbit/s	not run	153 Mbit/s (1020 retransmits)	97.5 Mbit/s (455 retransmits)

VPN values are receiver-side throughput. The two runs are shown separately and not averaged.

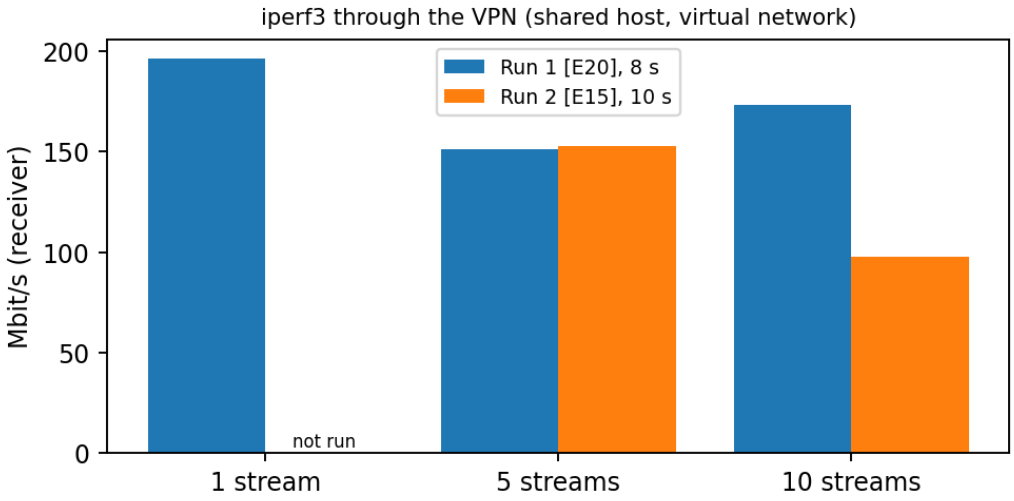


Figure 1. VPN throughput by number of streams, both runs [E20][E15]

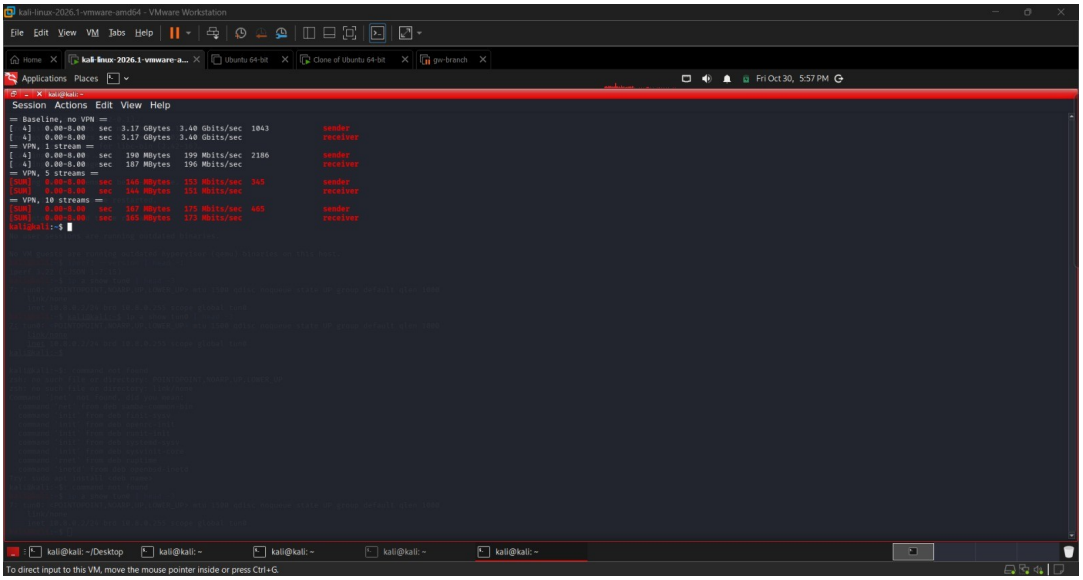


Figure 2. iperf3 run 1: baseline and VPN with 1, 5 and 10 streams [E20]

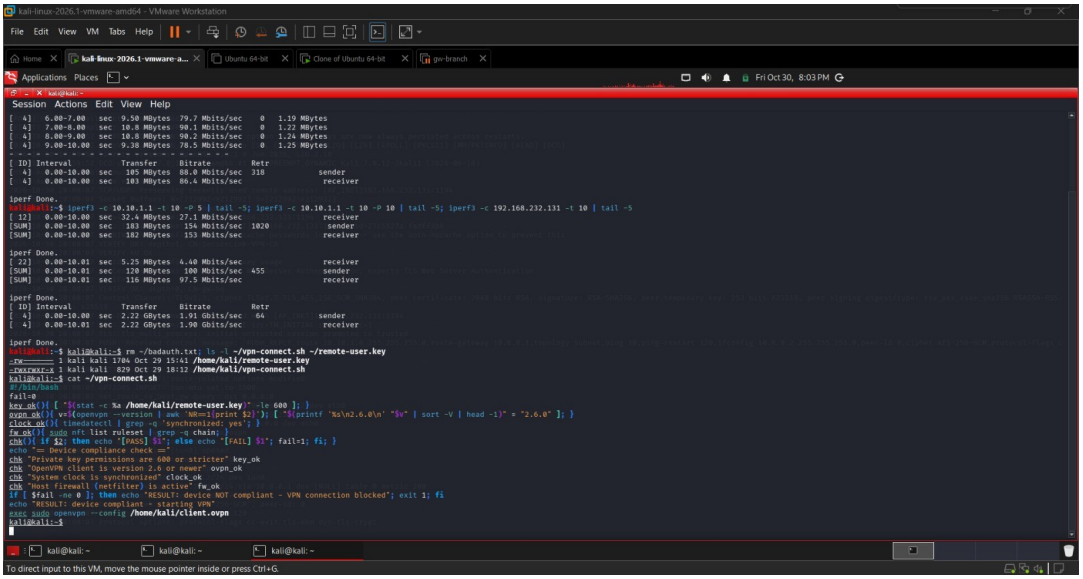


Figure 3. iperf3 run 2 (with vpn-connect.sh source) [E15]

3 Analysis

- VPN throughput was far below the no-VPN baseline in both runs, in the range of roughly 100 to 200 Mbit/s.
- Throughput did not scale with the number of streams; in run 2, 10 streams gave less than 5 streams.
- The baseline itself differed between runs (3.40 versus 1.90 Gbit/s), so the runs cannot be compared directly.
- Run 2 shows retransmits (1020 and 455). No cause has been established.
- CPU or virtualization overhead on the shared host is an untested hypothesis only.

4 Load balancing as built

Two OpenVPN instances run on one gateway (UDP 1194 with tun0, UDP 1195 with tun1) and the profile client-lb.ovpn connects to each [E21][E22]. Both instances were active [E16].

Aspect	Position
Who chooses the instance	The client
High availability	None: both instances share one gateway, so a gateway failure stops both
Traffic distribution	Not automatic and not measured
Status	Implemented but limited

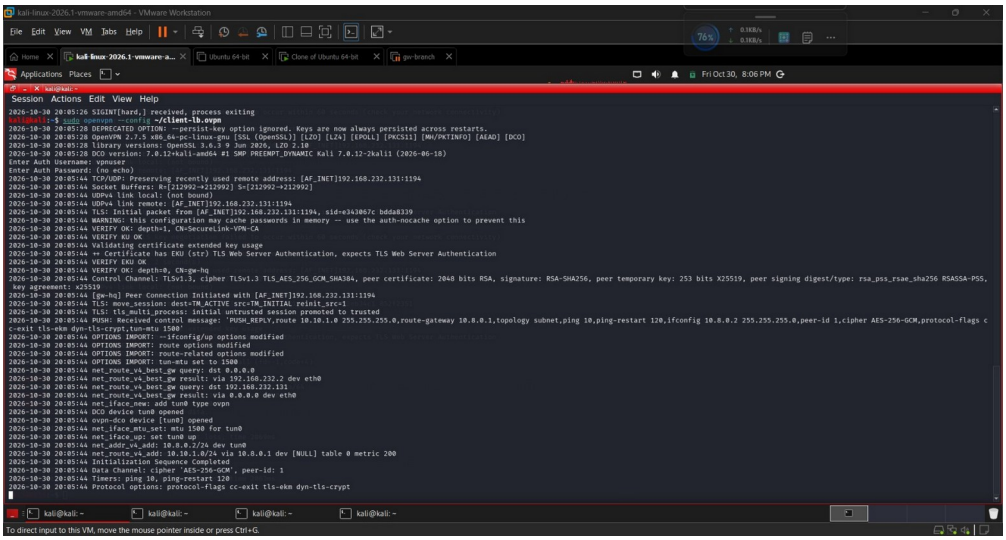


Figure 4. client-lb profile connected to instance 1, UDP 1194 [E21]

The connection to instance 2 (UDP 1195) is shown in [E22].

5 Scalability recommendations

All items below are recommendations, not implemented.

Area	Recommendation
Gateways	Several VPN gateways behind a real load balancer or DNS-based distribution, in more than one location
Identity	Per-user certificates and a central user directory (for example RADIUS) so users can be added and removed at scale
Address pools	Larger client pools and routes planned per gateway
Hardware	Test on dedicated hardware with AES acceleration before sizing
Monitoring	Track CPU, sessions and throughput per gateway

Capacity arithmetic (estimate, not measured): if one gateway sustained the lab's best VPN result of about 150 Mbit/s in total and an average user needed an assumed 2 Mbit/s, one gateway would serve roughly 75 concurrent users. The 2 Mbit/s figure is an assumption, and lab throughput may not carry over to production hardware.

6 Fuller test plan (not run)

- Many simultaneous clients with realistic traffic instead of one iperf3 client.
- Repeat runs on a quiet host with a fixed baseline, recording CPU on client and gateway.
- Compare UDP and TCP transport and different ciphers.
- Failover test with a second gateway.
- Long-duration soak test and tuning of retransmit sources.

7 Implementation status

Category	Items
Implemented and verified	iperf3 measurements with and without VPN [E15][E20]; both instances active [E16]
Implemented but limited	Two-instance setup on one gateway, client-chosen, no HA [E21][E22]; single-client load only
Recommended, not implemented	Real load balancing, multi-gateway scaling, capacity testing with many users